



Caught in a Martian dust storm

NASA's Mars rover, Opportunity, is battling a dust storm at the Mars Preservation Valley, since first detected on June 1. <https://digit.in/jul18-72-1>



Tennessee Woman sues NASA

A woman has sued NASA over a vial of moon dust which she claims to be a childhood gift from Neil Armstrong. <https://digit.in/jul18-72-2>

Also the Norm

The animal kingdom has come up with many approaches to making sure their genes are passed on to another generation.

Aditya Madanapalle |

Aditya@digit.in

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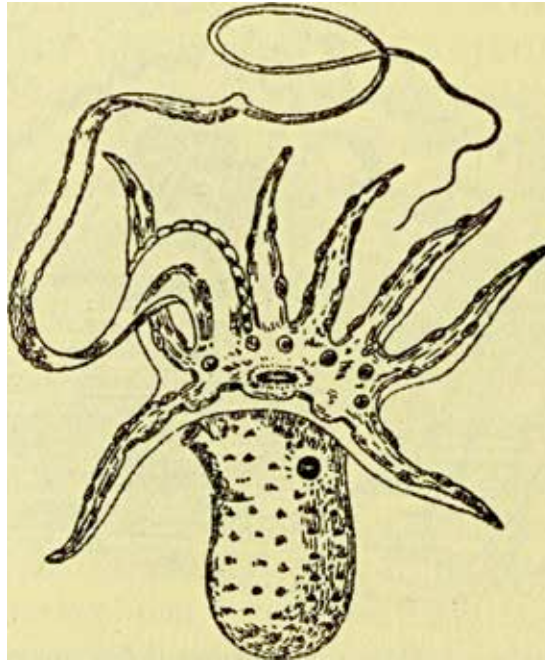
ost organisms want to produce as many offspring as possible, and make sure that these offspring have

the best chances of survival. The competition is with the opposite gender, as well as other adults. In this struggle, males and females of various species have evolved various strategies to optimise the chances of their genes passing on. However, it is not always so binary. At times a species may be made up entirely of females, and at other times the same animal might be a male and female at once. This makes for some complicated relationships.

PARTHENOGENESIS

Parthenogenesis is a process of reproduction where the embryo grows directly from an unfertilised egg. As a result, it is a form of asexual reproduction, where the species is entirely made up of females.

Perhaps the most intensely studied, and consequently the most well known example of a species that reproduces exclusively through parthenogenesis is the New Mexico whiptail lizard, or the *Cnemidophorus neomexicanus*. There are fifteen species in the genus *Cnemidophorus* that reproduce through parthenogenesis. The populations



The Argonaut octopus, the extra long limb is the hectocotylus

of the lizards have very little genetic diversity within a particular lineage. Even though the species are entirely made up of females, the animals continue to exhibit sexual behavior. The function is to stimulate the right hormones to optimise the process of reproduction. To do this, some females take on the role of males, and mount the females.

A few species of crabs, spiders and snails are also known to reproduce through parthenogenesis. There are some creatures that normally reproduce sexually, such as komodo dragons, boas or pythons, but have found to reproduce by parthenogenesis. Such cases are considered accidental parthenogenesis. There are some labs around the world experimenting in artificially inducing parthenogenesis in mammals. Some experiments involving rabbits and rats have been successful. However, parthenogenesis in the wild has not been observed with any mammalian species.

HERMAPHRODITES

Not all organisms are sexually differentiated. Some creatures have two sets of functional sex organs, one male and one female. These kinds of creatures are known as hermaphrodites. Some creatures may also change their genders depending on the conditions. These creatures are known as sequential hermaphrodites. There are two ways hermaphrodites can mate. Bilateral sperm transfer means that both the partners exchange the sperm required to reproduce. In unilateral sperm transfer, one creature inseminates

the other. The process of mating can even get violent.

Banana slugs have disproportionately large male sex organs. Despite the slugs having both sex organs, researchers noticed that an unusually large number of the animals were missing the penis. This is because after mating, the banana slugs have a tendency of biting off the penis of their partners. This is to reduce the competition, and a strategy to maximise the number of progeny for a single individual.

Some species of marine flatworms engage in a behavior known as penis fencing. Both the animals compete to be the one to inseminate the other. The organs used are sharp two headed probes, which attempt a traumatic insemination. This is known as intra-dermal hypodermic insemination. Essentially, the creatures attempt to stab their partners and inject the sperm directly into the circulatory system. A mating pair can either transfer the